

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_u^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_v^2$. We specifically wish to test the null hypothesis $\sigma_u^2 = \sigma_v^2$ against $\sigma_u^2 \neq \sigma_v^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_v^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | x_i, MARRIED = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(e_i | x_i, MARRIED = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

c. Following the regression in part (b), we carry out the NRC test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$\begin{array}{l} df \\ SSE_M \\ SSE_F \\ SSE_M \end{array} \begin{array}{l} 576 \\ 3 \\ 573 \end{array} \quad MSE_M = \frac{97161.9174}{573} = 169.569$$

$$\hat{\sigma}_M^2 = 13.0219 \quad NP = 1000 - 579 = 423$$

$$a. H_0: \frac{\sigma_M^2}{\sigma_F^2} = 1$$

$$H_1: \frac{\sigma_M^2}{\sigma_F^2} \neq 1$$

$$F = \frac{\frac{S_F^2 (n_F - 1)}{\sigma_F^2 (n_F - 1)}}{\frac{S_M^2 (n_M - 1)}{\sigma_M^2 (n_M - 1)}} \sim F(n_F - 1, n_M - 1)$$

$$RRE \Psi^* > F_{0.025}(422, 576) = 0.93$$

$$\Psi^* < F_{0.975}(422, 576) = 1.19$$

$$\Psi^* = \frac{S_F^2}{S_M^2} \times \frac{\sigma_M^2}{\sigma_F^2} = \frac{169.569}{12.024^2} \times 1 = 1.1736$$

$\therefore$ do not reject H_0 .
there is no evidence that $\sigma_M^2 / \sigma_F^2 \neq 1$

$$b. H_0: \frac{\sigma_S^2}{\sigma_M^2} \leq 1$$

$$H_1: \frac{\sigma_S^2}{\sigma_M^2} > 1$$

$$n_S = 400 \quad SSE = 56231.0382 \quad df \text{ of } SSE = 4$$

$$n_M = 600 \quad SSE = 100703.0471$$

$$MSE_S = \frac{56231.0382}{395} = 142.3571 = \hat{\sigma}_S^2$$

$$MSE_M = \frac{100703.0471}{595} = 169.2488 = \hat{\sigma}_M^2$$

$$F = \frac{\frac{S_M^2 (n_M - 1)}{\sigma_M^2 (n_M - 1)}}{\frac{S_S^2 (n_S - 1)}{\sigma_S^2 (n_S - 1)}} \sim F(399, 599)$$

$$RRE \Psi^* > F_{0.05}(399, 599) = 1.134$$

$$\Psi^* = \frac{S_M^2}{S_S^2} \times 1 = \frac{169.2488}{142.3571} = 1.19 \in RR$$

We shall reject H_0 , there is evidence that $\sigma_S^2 \neq \sigma_M^2$.

$$c. \text{Breusch Pagan} \quad H_0: \text{there is no heteroscedasticity.} \\ NR^2 = 59.03. \quad H_1: \text{there is heteroscedasticity.} \\ \chi^2_{0.95}(4) = 9.49$$

$$NR^2 \in RR$$

We should reject H_0 and conclude this model has heteroscedasticity and this result matches the one in problem (b).

$$d. \text{White test.}$$

$$\Psi^* = 78.82$$

$$df \text{ in white test}$$

$$F \text{ 始変数} = 4$$

$$平方 = 4$$

$$交互 = 6$$

$$\text{total } df = (4) \Rightarrow \chi^2_{0.95}(14) = 23.68$$

$$\Psi^* \in RR$$

We should reject H_0 and conclude this model has heteroscedasticity and this result matches the one in problem (b).

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se) (2.36) (0.14) (0.031) (1.05) (0.81)

robust SE (robse) (2.50) (0.16) (0.029) (0.84) (0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

f. in problem (b) tests if $\sigma^2_{\epsilon} = \sigma^2$
but in problem (f) with white test
is testing if the coefficient is significant.
so there is no conflict.

e. SE in intercept. ↑ (wider)
EDUC

SE in EXPER
METRO
FEMALE ↓ (narrower)

There is no inconsistency in this result
robust SE doesn't promise it to be bigger.

- 8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + \epsilon$$

MILES is miles driven per year. *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

a.

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

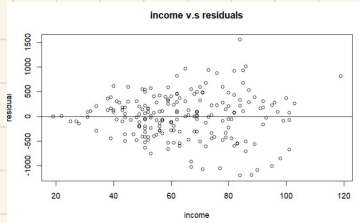
Residuals:
    Min       1Q   Median       3Q      Max
-1198.14  -295.31   17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775   -2.306  0.0221 *
income       14.201      1.800    7.889 2.10e-13 ***
age          15.741      3.757    4.189 4.29e-05 ***
kids        -81.826     27.130   -3.016  0.0029 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

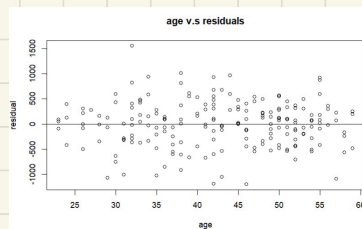
Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,    Adjusted R-squared:  0.3305
```

```
> confint(model, 'kids', level=0.95)
           2.5 %    97.5 %
kids -135.3298 -28.32302
```

b



the plot shows heteroscedasticity



plot is okay

$$H_0: \sigma^2_{\text{higher}} = \sigma^2_{\text{lower}} \Rightarrow \frac{\sigma^2_{\text{higher}}}{\sigma^2_{\text{lower}}} = 1$$

$$H_1: \sigma^2_{\text{higher}} \neq \sigma^2_{\text{lower}}$$

$$F = \frac{s^2_{\text{higher}}(n-1)}{s^2_{\text{lower}}(n-1)} \sim F(n-1, n-1)$$

$$RRE F^* < F_{0.975}(89, 89)$$

$$F > F_{0.025}(89, 89)$$

F-statistic = 3.104061
p-value = 1.64001e-07

- d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more mile on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 \text{INCOME}_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more mile on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

d. robust SE.

```

      Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.5480   142.6548  -2.7447  0.0066190 **
income       14.2013     1.9389    7.3246  6.083e-12 ***
age          15.7409     3.9657    3.9692  0.0001011 ***
kids         -81.8264    29.1544  -2.8067  0.0055112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

> cat('beta 95% CI', lower, higher)
beta 95% CI -138.969 -24.68383

```

beta = coefficient of kids.

SE of β_k has increased, compared to (a)'s model ^(-135, -28) and hence the 95% CI of coefficient of kids become wider.

$$e. \hat{\sigma}_i^2 = \frac{1}{\text{income}_i^2} \cdot \hat{\sigma}^2$$

↙ weight.

```

Coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept) -424.996   121.444  -3.500  0.000577 ***
income       13.947     1.481    9.420  < 2e-16 ***
age          16.717     3.025    5.527  1.03e-07 ***
kids         -76.806    21.848  -3.515  0.000545 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.765 on 196 degrees of freedom
Multiple R-squared:  0.4573,    Adjusted R-squared:  0.449
F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16

```

```

> confint(model_gls, 'kids', level=0.95)
      2.5 %      97.5 %
kids -119.8945 -33.71808

```

↑
conventional GLS



Compared to (d) & (a)
the CI is narrower.

We might say GLS can capture heteroscedasticity



$$\beta_k \in [-76.8063 \pm 1.96 \times 22.6186] \\ = [-121.1388, -32.4738]$$

```

t test of coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept) -424.9962    95.8035  -4.4361  1.526e-05 ***
income       13.9473     1.3470  10.3545  < 2.2e-16 ***
age          16.7175     2.7974   5.9761  1.061e-08 ***
kids         -76.8063    22.6186  -3.3957  0.0008286 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

↑
robust GLS.